

Swayam Prakash Panda

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Education

VIT Bhopal University

Expected May 2027

B.Tech in Computer Science and Engineering (AI & ML), CGPA: 8.63/10

Bhopal, India

- **Relevant Coursework:** Data Structures & Analysis of Algorithms, Applied Machine Learning, Artificial Neural Networks, Applied Linear Algebra, Probability & Statistics, Database Management Systems, Object-Oriented Programming, Cloud Computing

Experience

IIT Bombay

May 2026 – Present

Summer Intern (FOSSEE)

Mumbai, Maharashtra

- Engineered an interactive 3D plotting interface for the Osdag Plate Girder module using **Python** and **Plotly**.
- Integrated the **PySide/Qt** Output Dock with the plotting widget to enable real-time visualization of force combinations, reducing manual configuration steps for users by **60%**.

TechMaster AI Digi Pvt. Ltd.

Jan 2026 – Feb 2026

Python Developer Intern

Bhopal, Madhya Pradesh

- Built a **FastAPI**-based text moderation API using a dual-layer detection pipeline (dictionary + ML fallback) with **sub-1 ms** inference latency.
- Developed a real-time speech-to-text microservice using **Vosk** and **WebSockets** for low-latency streaming audio.
- Built a **JWT** + **Bcrypt** authentication backend for stateless session management and migrated frontend authentication from **Supabase** to **Firebase**.
- Contributed to a cloud migration across **Vercel**, **AWS**, and **Firebase** that reduced projected infrastructure costs by **45%**, and set up a **Docker**-based CI/CD workflow to streamline deployments.

Projects

True Random Number Generator using Biological Entropy | Python, OpenCV, NumPy, SHA-256, NIST Test Suite [\[Source Code\]](#)

- Developed a novel True Random Number Generator (TRNG) leveraging **biological entropy** from goldfish swimming patterns, implementing a computer vision pipeline for real-time centroid tracking.
- Used HSV color masking and Euclidean distance metrics to extract motion-based entropy and generate 256-bit **cryptographic** seeds.
- Validated statistical randomness via the **NIST Statistical Test Suite**, passing 13 of 14 benchmarks

Leptospirosis Outbreak Risk Predictor | Python, Scikit-learn, React, Node.js, PostgreSQL [\[Source Code\]](#) [\[Demo Link\]](#)

- Trained a **Random Forest regression** model on climate feature records across 26 European nations (2007 – 2023), achieving an R^2 score of **0.833** to predict disease outbreak risks.
- Built a data pipeline integrating **ECDC surveillance** and **ERA5 climate datasets**, and developed a full-stack web application with geospatial visualization of epidemiological risk factors.

Leadership & Achievements

- **Technical Leadership:** Student Coordinator for **GeeksforGeeks** (led 50-member team for campus-scale events) and Technical Lead at **AI Innovators Hub** (coordinated development of **Darzi AI**).
- **IEEE Research:** Published first-author research on **BERT-based Misinformation Portability** (RCSM 2025) and **Biological Entropy RNGs** using computer vision (ASSIC 2026).
- **Honors & Certifications:** Finalist at Johns Hopkins Health Hack 2025; 7th Place @ Startup Star 2022 (out of 300+ teams); Google Data Analytics Professional Certificate.

Technical Skills

Languages: Python, JavaScript, TypeScript, C++, SQL, HTML/CSS.

AI/ML & Data: Scikit-learn, Pandas, NumPy, OpenCV, BERT, RAG, LLM Pipelines, Vosk, Plotly, PySide/Qt.

Backend & Cloud: FastAPI, Node.js, Express.js, Docker, GitHub Actions, AWS, Vercel, Firebase, Cloudflare, Nginx.

Tools & Databases: Git, PostgreSQL, MongoDB, Postman, VS Code, Agile/Scrum.